-- 1. Create customers and orders Tables with a Foreign Key

```
CREATE TABLE customers (
   customer_id INT PRIMARY KEY,
   customer_name VARCHAR(100),
   city VARCHAR(50)
);
CREATE TABLE orders (
   order_id INT PRIMARY KEY,
   customer_id INT,
   product_name VARCHAR(100),
   amount DECIMAL(10,2),
   FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);
```

-- 2. Insert 5 Customers and 5 Orders

```
-- Customers
INSERT INTO customers VALUES
(1, 'Alice', 'Chennai'),
(2, 'Bob', 'Mumbai'),
(3, 'Charlie', 'Delhi'),
(4, 'David', 'Bangalore'),
(5, 'Emma', 'Hyderabad');
-- Orders
INSERT INTO orders VALUES
(101, 1, 'Laptop', 55000.00),
```

```
(102, 2, 'Mouse', 500.00),

(103, 1, 'Keyboard', 1200.00),

(104, 3, 'Monitor', 8000.00),

(105, 4, 'Printer', 6500.00);

-- 3. INNER JOIN: Fetch Customer Names with Their Orders

SELECT

   c.customer_name,

   o.order_id,

   o.product_name,

   o.amount

FROM customers c

INNER JOIN orders o

ON c.customer_id = o.customer_id;

-- --4. LEFT JOIN: Fetch All Customers Even Without Orders

SELECT

   c.customer_name,

   o.order_id,

   o.product_name

FROM customers c

LEFT JOIN orders o

ON c.customer_id = o.customer_id;

--  -- 5. Normalize a table with repeating groups into 2NF.

CREATE TABLE Customers (

   CustomerID INT PRIMARY KEY,

   CustomerName VARCHAR(100)

);
```

```
CREATE TABLE Orders (
   OrderID INT PRIMARY KEY,
   CustomerID INT,
   FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

CREATE TABLE OrderItems (
   OrderID INT,
   ProductName VARCHAR(100),
   PRIMARY KEY (OrderID, ProductName),
   FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
);
INSERT INTO Customers VALUES
(1, 'Alice');
INSERT INTO Orders VALUES
(101, 1);
INSERT INTO OrderItems VALUES
(101, 'Laptop'),
(101, 'Mouse'),
(101, 'Keyboard');
```

-- 6. Show an example of 3NF using sample data.

```
CREATE TABLE Cities (
   City VARCHAR(50) PRIMARY KEY,
   State VARCHAR(50)
);
```

```
CREATE TABLE Customers_3NF (
   CustomerID INT PRIMARY KEY,
   CustomerName VARCHAR(100),
   City VARCHAR(50),
   FOREIGN KEY (City) REFERENCES Cities(City)
);
INSERT INTO Cities VALUES
('Chennai', 'Tamil Nadu'),
('Mumbai', 'Maharashtra');

INSERT INTO Customers_3NF VALUES
(1, 'Alice', 'Chennai'),
(2, 'Bob', 'Mumbai');
SELECT c.CustomerID,
      c.CustomerName,
      c.City,
      ci.State
FROM Customers_3NF c
JOIN Cities ci
ON c.City = ci.City;
```